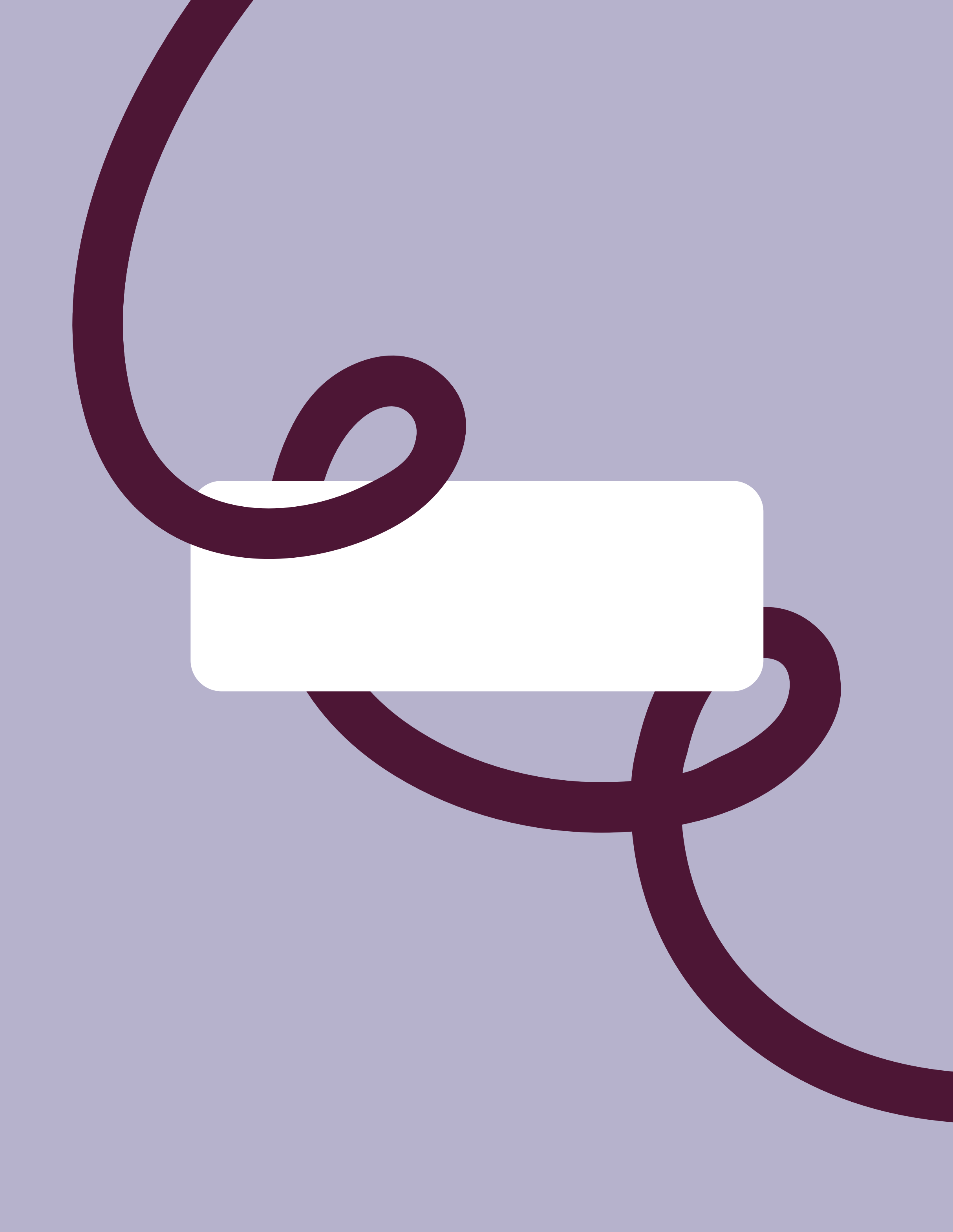
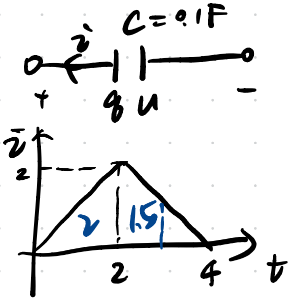


①


C=0.1F
i
+
q u
−
i
2
2
1.5
2
4
t

t=0s, $u_c(0)=-3.5V$

t=3s, $u_c=?$

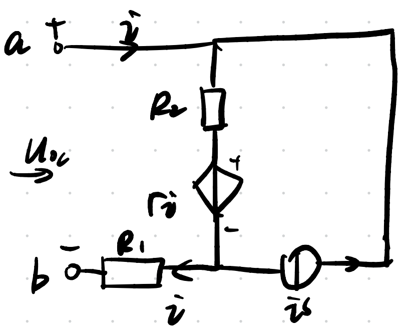
$Q=Cu$

$$u=\frac{1}{C}\int_{-\infty}^{3} i(y)\,dy$$

$$=\frac{1}{C}\int_{0}^{3} i(y)\,dy-3.5V$$

$$=-35V-3.5V=-38.5V$$

②

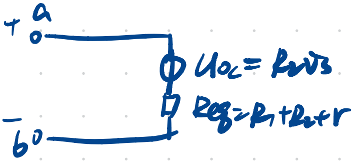

a
+
i
R_2
U_{oc}
+
ri
−
R_1
b
−
i
i_s

求戴维南，诺顿等效

① 求U_{oc} $i=0$. ☆ $U_{oc}=R_2 i_s$.

② 求R_{eq} $R_{eq}=\frac{(R_2+R_1+r)i}{i}=R_1+R_2+r$

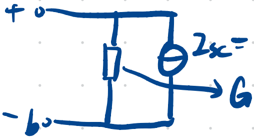
戴:



$U_{oc}=R_2 i_s$

$R_{eq}=R_1+R_2+r$

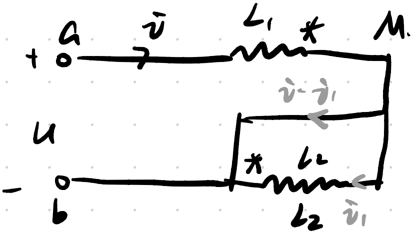
诺:



$I_{sc}=\frac{R_2 i_s}{R_1+R_2+r}$

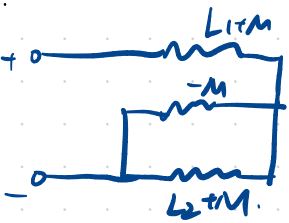
$G=\frac{1}{R_1+R_2+r}$

③


a
i
L_1
*
M.
+
u
−
b
$i-i_1$
*
L_2
L_2
i_1

求Leq.

⇒



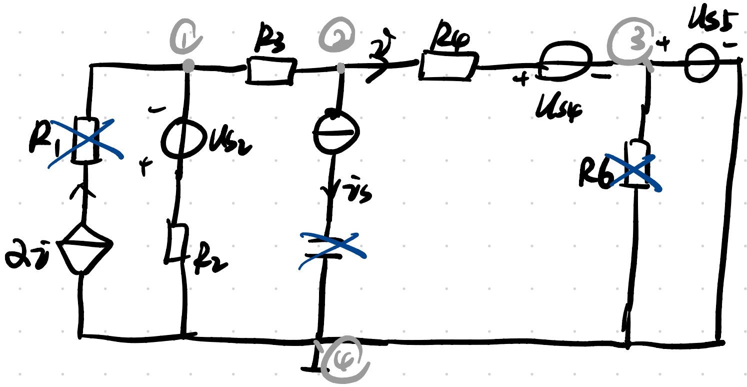
$$L_{eq}=L_1+M+\frac{(L_2+M)(-M)}{L_2+\cancel{M-M}}$$

$$=L_1+M-M-\frac{1}{L_2}M^2=L_1-\frac{1}{L_2}M^2$$

$$\varepsilon=L_1\frac{di}{dt}+M\frac{di_1}{dt}+\underbrace{L_2\frac{di_1}{dt}+M\frac{di}{dt}}_{=0}$$

$$=\left(L_1+M\cdot\left(-\frac{M}{L_2}\right)\right)\cdot\frac{di}{dt}\quad\Rightarrow L_{eq}=L_1-\frac{1}{L_2}M^2$$

④


①
R_3
②
i
R_4
+
−
U_{s4}
③
+
U_{s5}
−
R_1
$2i$
−
U_{s2}
+
R_2
i_s
R_6
④

列节点电压方程

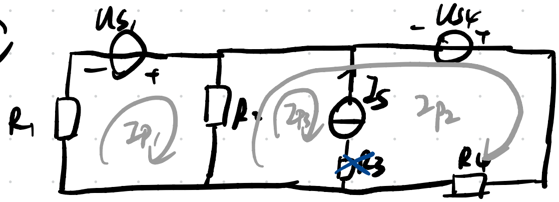
①: $\left(\frac{1}{R_2}+\frac{1}{R_3}\right)U_{n1}-\frac{1}{R_3}\cdot U_{n2}=2i-U_{s2}$

②: $\left(\frac{1}{R_3}+\frac{1}{R_4}\right)U_{n2}-\frac{1}{R_3}U_{n1}-\frac{1}{R_4}U_{n3}=\frac{U_{s4}}{R_4}-i_s$

③: $U_{n3}=U_{s5}$

$i=\frac{1}{R_4}\cdot(U_{n2}-U_{n3}-U_{s4})$

⑤



列回路电流方程

$$\begin{cases} U_{S1} = (I_{p1}-I_{p2}-I_{p3})R_2 + I_{p1}R_1 \\ I_{p3} = -I_S \\ U_{S4} = I_{p2}R_4 + (I_{p2}+I_{p3}-I_{p1})R_2 \end{cases}$$

⑥ 工频. $S_N = 100k\ V\cdot A$. $U_N = 220V$. 白炽: $P=40W$. 日光: $P=10W$ $\lambda = 0.4$

(i) 一个白炽+1个日光为一组. 最多 N组? 求此时I? $\lambda=1$

$$\lambda = \cos\theta = \frac{2}{5} \Rightarrow \tan\theta = \frac{\sqrt{21}}{2}.$$

$$\Rightarrow s = P\cdot\tan\theta = 5\sqrt{21}\ Var. \Rightarrow S_{总} = \sqrt{(10W+40W)^2 + s^2}$$

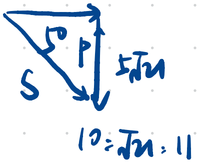
$$= 55\ VA$$

$$\Rightarrow n = \left\lfloor \frac{S_N}{S_{总}} \right\rfloor = \frac{1\times10^5\cdot VA}{55\cdot VA} = 1818 \quad I = \frac{S_N}{U_N} = 454.54A.$$

(ii) 并联 -||- 使 $\lambda' = 98\%$. 求 C ? 此时 I ? $\omega = 2\pi f = 100\pi\ rad/s.$

$$\delta Q = 1\times10^5 VA\cdot\frac{10}{11}\cdot\left(\frac{\sqrt{21}}{10} - \frac{\sqrt{1-(0.98)^2}}{0.98}\right) = 23199.90\ Var$$

$$= \omega C U^2 \Rightarrow C = \frac{\delta Q}{\omega U^2} = 1.53\times10^{-3}F.$$

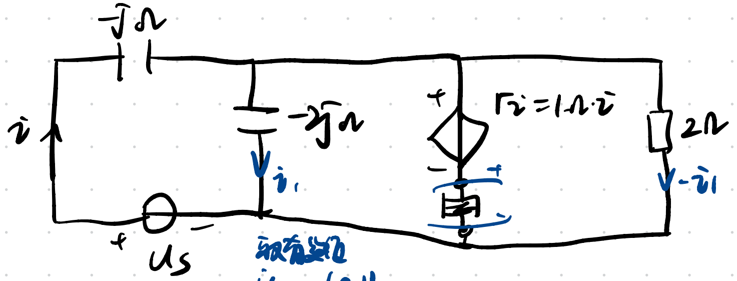


$$I = \frac{1\times10^5 VA\cdot\frac{10}{11}\cdot\frac{100}{98}}{220V} = 421.66A.$$

(iii) 再接入白炽.. 最多几盏？

$$W=\left\lfloor\frac{-1\times10^5 V\cdot A\cdot\frac{10}{11}\cdot\frac{100}{98}+1\times10^5\cdot V\cdot A}{40W}\right\rfloor = 180 \text{ 盏}$$

⑦ $u_s = 60\sqrt{2}\cos\omega t$ 求外接负载Z 改变时. 何时取 $P_{max}=?$


$-j\Omega$
$\dot I$
$-2j\Omega$
$\dot I_1$
+
−
$r\dot I = 1\Omega\cdot\dot I$
+
2Ω
$-\dot I_1$
+
−
u_s

取有效值
$\dot U_s = 60V.$

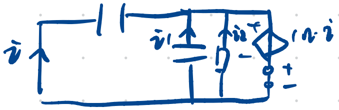
① 求U_{oc}. ⇒ $\dot I = \frac{U_s}{-j+\frac{-4j}{2-2j}} = \frac{U_s}{1-2j} = U_s\cdot\frac{1+2j}{5}$

取有效值！

$=(24-12j)V.$

⇒ $U_{oc} = U_s + j\dot I - 1\cdot\dot I = U_s\cdot(1-\frac{1+2j}{5}\cdot(1-j)) = U_s\cdot(\frac{2}{5}-\frac{1}{5}j)$

② 求 Z_{eq}.


$\dot I$
$\dot I_1$
$\dot I_2$
+
−
$1\Omega\cdot\dot I$
+
−

$-j\dot I = -2j\dot I_1 = 2\dot I_2$

$$Z_{eq} = \frac{j\Omega\dot I - 1\Omega\cdot\dot I}{-(\dot I+\dot I_1+\dot I_2)} = \frac{-j+1}{1+\frac{1}{2}-\frac{j}{2}}$$

$$=\frac{2(1-j)(3+j)}{(3-j)(3+j)} = \frac{1}{5}(4-2j)$$

⇒ $Z^* = 0.8+0.4j.$

$$P_{max} = \frac{(24^2+12^2)}{4\times0.8} = 225W.$$